HDOC Day-18

Menu-Driven C Program: Digit-Based Operations

Aim

To write a menu-driven C program using switch-case and separate void user-defined functions to check whether a positive integer is a Strong, Armstrong, Harshad, or Duck number.

Algorithm

1. Start the program.

2. Declare four void functions: strong(), armstrong(), harshad(), and duck().

3. Display the menu and read the user choice.

4. Use switch-case to call the required function.

5. Inside the selected function, read the positive integer.

6. Perform the required digit-based calculation and display the result.

7. Stop the program.

C Program

```
#include <stdio.h>

void strong();
void armstrong();
void harshad();
void duck();

int main()
{
    int ch;

    printf("1. Strong Number\n");
    printf("2. Armstrong Number\n");
    printf("3. Harshad Number\n");
    printf("4. Duck Number\n");
    printf("Enter choice: ");
    scanf("%d", &ch);

    switch(ch)
    {
        case 1: strong(); break;
        case 2: armstrong(); break;
        case 3: harshad(); break;
        case 4: duck(); break;
        default: printf("Invalid choice");
    }
    return 0;
}

void strong()
{
```

```c
    int n, temp, d, sum = 0, fact, i;
    printf("Enter number: ");
    scanf("%d", &n);
    temp = n;

    while(temp > 0)
    {
        d = temp % 10;
        fact = 1;
        for(i = 1; i <= d; i++)
            fact = fact * i;
        sum = sum + fact;
        temp = temp / 10;
    }

    if(sum == n)
        printf("Strong Number");
    else
        printf("Not a Strong Number");
}

void armstrong()
{
    int n, temp, d, sum = 0;
    printf("Enter number: ");
    scanf("%d", &n);
    temp = n;

    while(temp > 0)
    {
        d = temp % 10;
        sum = sum + d*d*d;
        temp = temp / 10;
    }

    if(sum == n)
        printf("Armstrong Number");
    else
        printf("Not an Armstrong Number");
}

void harshad()
{
    int n, temp, d, sum = 0;
    printf("Enter number: ");
    scanf("%d", &n);
    temp = n;

    while(temp > 0)
    {
        d = temp % 10;
        sum = sum + d;
        temp = temp / 10;
    }

    if(n % sum == 0)
        printf("Harshad Number");
    else
        printf("Not a Harshad Number");
}

void duck()
{
    int n, d, flag = 0;
    printf("Enter number: ");
    scanf("%d", &n);

    while(n > 0)
```

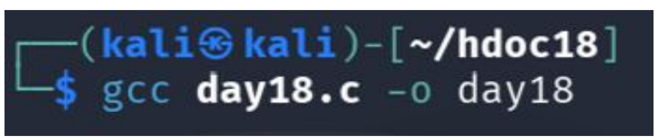
```
    {
        d = n % 10;
        if(d == 0)
            flag = 1;
        n = n / 10;
    }

    if(flag == 1)
        printf("Duck Number");
    else
        printf("Not a Duck Number");
}
```

Test Case Table

Choice	Input	Expected Result	Operation
1	145	Strong Number	Strong
1	123	Not a Strong Number	Strong
2	153	Armstrong Number	Armstrong
2	123	Not an Armstrong Number	Armstrong
3	18	Harshad Number	Harshad
3	19	Not a Harshad Number	Harshad
4	102	Duck Number	Duck
4	1234	Not a Duck Number	Duck

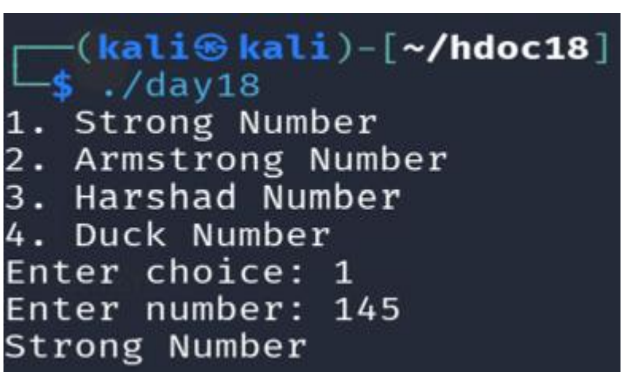
Compilation

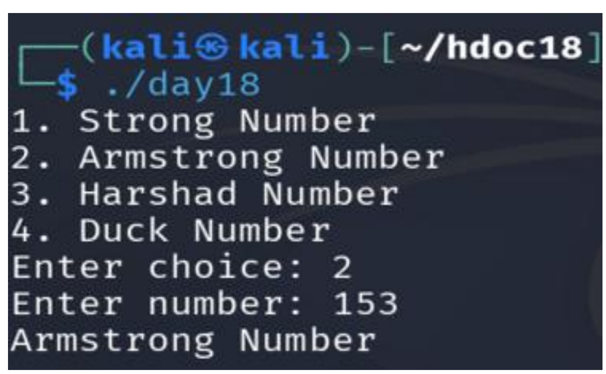
`gcc day18.c -o day18`



Execution

`./day18`





```
┌──(kali㉿kali)-[~/hdoc18]
└─$ ./day18
1. Strong Number
2. Armstrong Number
3. Harshad Number
4. Duck Number
Enter choice: 3
Enter number: 18
Harshad Number
```

```
┌──(kali㉿kali)-[~/hdoc18]
└─$ ./day18
1. Strong Number
2. Armstrong Number
3. Harshad Number
4. Duck Number
Enter choice: 4
Enter number: 102
Duck Number
```